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CSA1603 CO3 - AT2 DWDM

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Slot C

1. A shepherd boy gets bored tending the town's flock. To have some fun he cries out, "wolf!" even though no wolf is in sight. The villagers run to protect the flock, but then get really mad when they realize the boy was playing a joke on them. It also predicts doors of a cat set into lamb chops. The towns goes hungry. panic ensues. Identify TP, FP, TN, FN

Act \ pre	Y	N
Y	TP	FN
N	FP	TN

* positive = wolf is actually present
* Negative = No wolf is present
* prediction = Boy cries "wolf!"

* TP : Real wolf appears + boy cries "wolf"
* FP : No wolf + boy falsely cries "wolf" →
* FN : Real wolf appears + no effective "wolf"
* TN : No wolf + no false warning → TN

2. Assume there are 100 images. 30 of them contain a cat, the rest do not. A Machine learning Model predicts the occurrence of a cat in 25 of 30 cat images. It also predict absence of cat in 50 of the 70 no cat image. Identify TP, FP, FN, TN & calculate the precision, recall & F1 score.

	act cat	act no cat	
pred cat	TP=25	FP=20	45
pred no cat	FN 5	TN=50	55
Total	30	70	100

* Total image = 100
* cat image = 30

* No cat image = 70
* Model correctly predicts cat in 25 of 30 cat image
* Model correctly predicts no cat in 50 of 70 no cat image

∴,

* TP = 25 → correctly identified cat
* FP = 20 → predict cat, but actually no cat
* FN = 5 → predicted no cat, but actually cat
* TN = 50 → correctly identify no cat

<u>Precision</u>;

$$= \frac{TP}{TP+FP}$$

$$= \frac{25}{25+20}$$

$$= \frac{25}{40}$$

$$= 0.556$$

$$= 55.56\%$$

Recall ;

$$= \frac{TP}{TP+FN}$$

$$= \frac{25}{25+5}$$

$$= \frac{25}{30}$$

$= 0.833$

$= 83.33\%$

F1 score ;

$$F_1 = 2 \times \frac{\text{precision} \times \text{Recall}}{\text{precision} + \text{Recall}}$$

$$= 2 \times \frac{0.556 \times 0.833}{0.556 + 0.833}$$

$= 0.667$

3. An email service provider wants to classify email as spam or not spam using features

* contain "offer" (yes/No) * Length (short/long)
* contain "free" (yes/No)

1. calculate prior probabilities :

* P(spam) , P(Not spam)

2. Compute conditional probabilities for each feature

3. using Native Bayes, classify ;

* (offer = yes , Free = No , length = short)

4. Compare results using WEKA

There are 6 emails ;

* spam = 3
* Not spam = 3

1. prior probabilities ;

P(spam)

$$= \frac{3}{6} = 0.5$$

P(Not spam)

$$= \frac{3}{6} = 0.5$$

2. Conditional probabilities:

There are 3 spam;

Feature	value	count	Probability
offer	yes	2	2/3 = 0.66
offer	no	1	1/3 = 0.33
Free	yes	3	3/3 = 1.00
Free	no	0	0/3 = 0.00
length	short	2	2/3 = 0.66
length	length	1	1/3 = 0.33

$$P(\text{offer} = \text{yes} \mid \text{spam}) = \frac{2}{3}$$

$$P(\text{Free} = \text{no} \mid \text{spam}) = \frac{0}{3}$$

$$P(\text{length} = \text{short} \mid \text{spam}) = \frac{2}{3}$$

There are 3 no spam;

Feature	value	count	probability
offer	yes	1	1/3 = 0.33
offer	no	2	2/3 = 0.66
Free	yes	0	0/3 = 0.00
Free	no	3	3/3 = 1.00
length	short	1	1/3 = 0.33
length	long	2	2/3 = 0.66

3. Classify the new Email;

Offer = yes ; Free = No ; length = short

$$P(A/B) = \frac{P(B/A) \times P(A)}{P(B)}$$

$$\text{Spam} = 0.5 \times \frac{2}{3} \times 0 \times \frac{2}{3}$$

$$= 0$$

$$\text{Not spam} = 0.5 \times \frac{1}{3} \times 1 \times \frac{1}{3}$$

$$= \frac{1}{18}$$

$$= 0.0556$$

4. WEKA;

Naive Bayes classifier

Attribute	class spam	not spam
Offer		
yes	3.0	2.0
no	2.0	3.0
[Total]	5.0	5.0
Free		
yes	4.0	1.0
no	1.0	4.0
[Total]	5.0	5.0
Length		
short	3.0	2.0
long	2.0	3.0
[Total]	5.0	5.0

Ahsani)

Graph: